

Solve Any Data Analysis Problem

Chapter 5 - Project 4 - Example solution - Part 1 (alternative)

We need to find and extract:

- annual admissions breakdown
- breakdown of genre
- breakdown of distributors

Figure out where relevant tables are for a few years (2018 onwards so we have both pre- and "post-" COVID periods)

Process to install the `camelot` library:

- install Ghostscript (<https://camelot-py.readthedocs.io/en/master/user/install-deps.html#install-deps>)
- install Tkinter if not already installed (<https://www.activestate.com/products/tcl/>)
- install `camelot` (at time of writing, one of its dependencies currently requires Python 3.9 so is incompatible with newer versions)

```
In [1]: import numpy as np
import pandas as pd

import camelot

from IPython.display import display
```

Let's create a reusable function to extract and print all the tables in a PDF

```
In [2]: def extract_tables(pdf_path, pages="all", print_tables=True):
        """
        Extract all tables found in a PDF.

        `pdf_path`: file path pointing to the PDF
        `pages`: the page number(s) to read
        `print_tables`: whether to also print out all the tables that are found (default

        returns: a list of pandas DataFrames
        """

        print(f"Reading {pdf_path}")

        tables = camelot.read_pdf(pdf_path, pages=pages, flavor="stream")

        print(f"{tables.n} tables found.")
```

```

if tables.n > 0:
    dataframes = [table.df for table in tables]

    if print_tables:
        for index, df in enumerate(dataframes):
            print(f"\n#####\n\tTable {index}\n#####")
            display(df)

    return dataframes
return []

```

2018

For 2018 all necessary tables are in the same PDF so let's extract them all here.

```
In [3]: tables_2018 = extract_tables("./files/2019 - BFI yearbook 2019 - 888.pdf", pages="1
```

```

Reading ./files/2019 - BFI yearbook 2019 - 888.pdf
4 tables found.

```

```

#####
      Table 0
#####

```

	0	1	2	3
0	Table 2 Monthly UK cinema admissions, 2017 and...			
1	Month	2017 (million)	2018 (million)	% +/- on 2017
2	January	15.0	16.2	8.0
3	February	16.5	16.1	-2.3
4	March	16.2	13.5	-16.2
5	April	15.6	15.5	-0.9
6	May	11.3	13.7	21.0
7	June	9.6	10.4	8.6
8	July	17.8	15.6	-12.3
9	August	14.5	19.2	32.8
10	September	10.8	10.1	-6.0
11	October	12.1	16.0	32.6
12	November	14.1	14.8	5.4
13	December	17.2	15.7	-8.7
14	Total	170.6	177.0	3.7
15	Source: CAA, comScore			

Table 1
#####

	0	1	2	3	4	5
0	Table 17 Films released in the UK and Republic...					
1	Genre	Number of	% of	Gross box	% of total	Top performing title
2		releases	releases	office	box office	
3				(£ million)		
4	Action	77	9.8	361.3	27.7	Avengers: Infinity War
5	Animation	46	5.8	242.2	18.6	Incredibles 2
6	Drama	246	31.3	154.1	11.8	A Star Is Born
7	Comedy	144	18.3	88.3	6.8	Johnny English Strikes Again
8	Biopic	5	0.6	78.5	6.0	Bohemian Rhapsody
9	Musical	5	0.6	67.0	5.1	Mamma Mia! Here We Go Again
10	Family	10	1.3	60.3	4.6	Mary Poppins Returns
11	Fantasy	4	0.5	57.6	4.4	Fantastic Beasts: The Crimes of Grindelwald
12	Horror	38	4.8	57.1	4.4	A Quiet Place
13	Adventure	10	1.3	36.4	2.8	Ready Player One
14	Sci-fi	14	1.8	30.8	2.4	Solo: A Star Wars Story
15	Thriller	42	5.3	30.6	2.3	The Equalizer 2
16	Crime	10	1.3	17.9	1.4	Widows
17	Romance	17	2.2	11.8	0.9	The Guernsey Literary and Potato Peel Pie Society
18	Documentary	112	14.2	9.3	0.7	Free Solo
19	War	3	0.4	1.3	0.1	Journey's End
20	Western	2	0.3	0.1	<0.1	Sweet Country
21	Mystery	2	0.3	0.1	<0.1	Dark River
22	Total	787	100.0	1,304.8	100.0	
23	Source: comScore, BFI RSU analysis					

#####

Table 2

#####

	0	1
0	DISTRIBUTORS	cinemas into 2018. The top earning distributor...
1		Disney, which handled seven of the top 20 rele...
2	The top 10 feature film distributors had a 95...	2018 including Avengers: Infinity War, Incredi...
3	the market in 2018 from the release of 251 tit...	Black Panther, as well as Star Wars: The Last ...
4	all films on release). The same 10 distributor...	December 2017 release which continued to generate
5	market share of 96% in 2017 from 284 releases ...	significant box office in 2018. (The 24% box o...
6	releases). A further 128 film distributors wer...	generated by Walt Disney in 2018 is the highes...
7	the theatrical release of films in the UK and ...	achieved by a single distributor since our rec...
8	Ireland in 2018, compared with 117 companies o...	The second highest earning distributor was Uni...
9	top 10 in 2017. These distributors handled a t...	which released four of 2018's top 20 titles in...
10	titles (74% of all releases) but generated onl...	Mamma Mia! Here We Go Again, Jurassic World: F...
11	total box office.	Kingdom and The Grinch.
12	Table 1 shows box office takings by distributo...	The top earning independent distributor in 201...
13	on release during 2018, and hence includes rev...	Films, whose highest grossing titles were The ...
14	titles which were released in 2017 but remaine...	House With a Clock in Its Walls and Finding Yo...

#####

Table 3

#####

	0	1	2	3
0	on release during 2018, and hence includes rev...		Films, whose highest grossing titles were The ...	
1	titles which were released in 2017 but remaine...		House With a Clock in Its Walls and Finding Yo...	
2	Table 1 Distributor share of box office, UK an...			
3	Distributor	Market share	Films on release	Box office gross
4		(%)	in 2018	(£ million)
5	Walt Disney	23.6	24	325.6
6	Universal	19.5	40	268.5
7	20th Century Fox	14.5	28	199.3
8	Warner Bros	13.9	31	191.4
9	Sony	10.7	33	146.8
10	Paramount	4.8	12	66.0
11	eOne Films	3.2	22	43.9
12	StudioCanal	2.8	31	38.4
13	Lionsgate	1.5	21	21.2
14	Entertainment	1.1	9	15.6
15	Top 10 total	95.5	251	1,316.8
16	Others (128 distributors)	4.5	715	61.4
17	Total	100.0	966	2,695.0
18	Source: comScore			

Looks like the first table is the monthly breakdown we're looking for.

Mention insertion of Year everywhere

```
In [4]: admissions_2018 = (
    tables_2018[0]
    .loc[2:13,[0, 2]] # drop first two and last two rows, keep month and 2018 admis
)

admissions_2018.columns = ["Month", "Admissions (million)"]

admissions_2018.insert(0, "Year", 2018)

admissions_2018.head()
```

```
Out[4]:
```

	Year	Month	Admissions (million)
2	2018	January	16.2
3	2018	February	16.1
4	2018	March	13.5
5	2018	April	15.5
6	2018	May	13.7

```
In [5]: admissions_2018.tail()
```

```
Out[5]:
```

	Year	Month	Admissions (million)
9	2018	August	19.2
10	2018	September	10.1
11	2018	October	16.0
12	2018	November	14.8
13	2018	December	15.7

Genres (2018)

```
In [6]: genres_2018 = (
    tables_2018[1]
    .loc[4:21] # drop first 4 and last 2 rows
    .drop(columns=[2, 4])
)

genres_2018.insert(0, "Year", 2018)

genres_2018.columns = ["Year", "Genre", "Number of releases", "Gross box office (£
genres_2018.head()
```

```
Out[6]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
4	2018	Action	77	361.3	Avengers: Infinity War
5	2018	Animation	46	242.2	Incredibles 2
6	2018	Drama	246	154.1	A Star Is Born
7	2018	Comedy	144	88.3	Johnny English Strikes Again
8	2018	Biopic	5	78.5	Bohemian Rhapsody

```
In [7]: genres_2018.tail()
```

Out[7]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
17	2018	Romance	17	11.8	The Guernsey Literary and Potato Peel Pie Society
18	2018	Documentary	112	9.3	Free Solo
19	2018	War	3	1.3	Journey's End
20	2018	Western	2	0.1	Sweet Country
21	2018	Mystery	2	0.1	Dark River

Distributors (2018)

In [8]:

```
distributors_2018 = (
    tables_2018[3]
    .loc[5:] # drop first 5 rows
    .drop(index=[15, 17, 18]) # drop rows relating to totals EXCEPT the row totalli
)

distributors_2018.insert(0, "Year", 2018)

distributors_2018.columns = ["Year", "Distributor", "Market share", "Films on relea
distributors_2018.head()
```

Out[8]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
5	2018	Walt Disney	23.6	24	325.6
6	2018	Universal	19.5	40	268.5
7	2018	20th Century Fox	14.5	28	199.3
8	2018	Warner Bros	13.9	31	191.4
9	2018	Sony	10.7	33	146.8

In [9]:

```
distributors_2018.tail()
```

Out[9]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
11	2018	eOne Films	3.2	22	43.9
12	2018	StudioCanal	2.8	31	38.4
13	2018	Lionsgate	1.5	21	21.2
14	2018	Entertainment	1.1	9	15.6
16	2018	Others (128 distributors)	4.5	715	61.4

2019

```
In [10]: tables_2019 = extract_tables("./files/2020 - BFI Statistical Yearbook 2020 - 12815.
```

Reading ./files/2020 - BFI Statistical Yearbook 2020 - 12815.pdf
5 tables found.

```
#####  
Table 0  
#####
```

	0	1	2	3
0	Table 1 Admissions in selected global territor...			
1	Territory	Admissions 2018	Admissions 2019	+/- 2018 (%)
2		(million)	(million)	
3	China	1,717	1,727	0.6
4	India	1,463	1,592	8.8
5	USA and Canada	1,310	1,256	-4.1
6	Mexico	332	350	5.5
7	South Korea	216	227	4.8
8	Russia	200	219	9.5
9	France	201	213	6.0
10	Japan	169	195	15.2
11	UK	177	176	-0.5
12	Brazil	163	172	5.6
13	Indonesia	135	146	7.8
14	Germany	105	119	12.6
15	Italy	92	105	14.2
16	Spain	99	105	4.8
17	Poland	60	61	1.5
18	Turkey	70	59	-15.6
19	Source: CAA, BFI, European Audiovisual Observa...			

```
#####  
Table 1  
#####
```

	0	1	2	3
0	Table 2 Monthly UK cinema admissions, 2018 and...			
1	Month	2018 (million)	2019 (million)	% +/- on 2018
2	January	16.2	13.7	-15.4
3	February	16.1	12.2	-24.0
4	March	13.5	11.4	-15.5
5	April	15.5	16.0	3.1
6	May	13.7	16.6	20.8
7	June	10.4	13.9	33.7
8	July	15.6	18.7	19.5
9	August	19.2	15.6	-19.1
10	September	10.1	11.0	8.2
11	October	16.1	16.3	1.8
12	November	14.9	12.2	-17.8
13	December	15.7	18.5	18.1
14	Total	177.0	176.1	-0.5
15	Source: CAA, Comscore			

#####

Table 2

#####

0	1	2	3	4	5	
0	Souvenir and Gwen.					
1	Table 10 Films released in the UK and Republic...					
2	Genre	Number of	% of	Gross	% of total	Top performing title
3		releases	releases	box office	box office	
4				(£ million)		
5	Action	102	13.4	328.3	25.2	Avengers: Endgame
6	Animation	40	5.2	312.1	24.0	The Lion King
7	Drama	264	34.6	166.4	12.8	Downton Abbey
8	Comedy	136	17.8	83.5	6.4	Yesterday
9	Biopic	19	2.5	80.7	6.2	Rocketman
10	Thriller	20	2.6	75.3	5.8	Joker
11	Sci-fi	3	0.4	64.3	4.9	Star Wars: The Rise of Skywalker
12	Horror	37	4.8	60.6	4.7	It Chapter Two
13	Adventure	8	1.0	44.5	3.4	Aladdin
14	Family	5	0.7	42.7	3.3	Dumbo
15	Romance	7	0.9	18.9	1.5	Last Christmas
16	Musical	8	1.0	12.3	0.9	Cats
17	Documentary	99	13.0	7.8	0.6	Apollo 11
18	Crime	11	1.4	2.8	0.2	21 Bridges
19	Mystery	1	0.1	0.5	<0.1	The Souvenir
20	War	2	0.3	0.1	<0.1	Saving Private Ryan (D-Day 75th Anniversary)
21	Fantasy	2	0.3	<0.1	<0.1	Gwen
22	Total	764	100.0	1,300.9	100.0	
23	Source: Comscore, BFI RSU analysis					

Table 3
#####

0	DISTRIBUTORS	box office. Walt Disney's market share is the ...
1		achieved by a single distributor since our rec...
2	The top 10 feature film distributors in 2019 h...	added to the total from 20th Century Fox, whic...
3	of the market from the release of 248 titles (...)	acquired in early 2019, the total share would ...
4	on release). The top 10 distributors in 2018 a...	Disney titles made up eight of the top 20 rele...
5	share from 251 releases (26% of films on relea...	including the year's three highest earning fil...
6	130 film distributors were involved in the the...	Endgame (£89 million), The Lion King (£76 mill...
7	films in the UK and Republic of Ireland in 201...	Story 4 (£66 million). The second highest earn...
8	with 128 companies outside the top 10 in 2018....	in 2019 was Universal, which released four of ...
9	distributors outside the top 10 handled a tota...	20 titles including Downton Abbey (£28 million...
10	(73% of all films on release) but generated on...	Furious: Hobbs & Shaw (£21 million) and How to...
11	total box office.	Dragon: The Hidden World (£20 million).
12	Table 1 shows box office takings by distributo...	The top earning independent distributor in 201...
13	on release during 2019, and hence includes rev...	Lionsgate, whose highest grossing titles were ...
14	titles which were released in 2018 but remaine...	(£13 million), John Wick: Chapter 3 – Parabell...
15	into 2019. The top earning distributor was Wal...	and Angel Has Fallen (£7.5 million).
16	total revenues of £507 million, a 38% share of...	
17	Table 1 Distributor share of box office, UK an...	

 Table 4
 #####

	0	1	2	3
0	total revenues of £507 million, a 38% share of...			
1	Table 1 Distributor share of box office, UK an...			
2	Distributor	Market share	Films on release	Box office gross
3		(%)	in 2019	(£ million)
4	Walt Disney	37.9	24	506.8
5	Universal	13.9	39	185.6
6	Warner Bros	12.4	40	166.3
7	Sony	9.3	27	124.7
8	Paramount	5.9	13	79.0
9	20th Century Fox*	5.7	30	75.5
10	Lionsgate	4.2	19	56.4
11	Entertainment One	3.8	27	50.4
12	STX Entertainment	1.2	7	15.4
13	StudioCanal	1.0	22	14.0
14	Top 10	95.4	248	1,274.1
15	Others (Total 130 distributors)	4.6	671	61.6
16	Total	100.0	919	1,335.8
17	Source: Comscore			

Here it looks like admissions is table 1.

Technically we could have extracted the 2018 figures from here as well. If we wanted a genre or distributor-level breakdown for 2018, we would have needed to open those PDFs anyway, so it doesn't make a difference.

If you were being really careful, you could also compare these figures with the ones in the 2018 PDF to make sure they match...

```
In [11]: admissions_2019 = (
    tables_2019[1]
    .loc[2:13,[0, 2]] # ignore first two rows, last 2 rows, and keep first and third
)

admissions_2019.columns = ["Month", "Admissions (million)"]
admissions_2019.insert(0, "Year", 2019)
```

```
admissions_2019.head()
```

```
Out[11]:
```

	Year	Month	Admissions (million)
2	2019	January	13.7
3	2019	February	12.2
4	2019	March	11.4
5	2019	April	16.0
6	2019	May	16.6

```
In [12]: admissions_2019.tail()
```

```
Out[12]:
```

	Year	Month	Admissions (million)
9	2019	August	15.6
10	2019	September	11.0
11	2019	October	16.3
12	2019	November	12.2
13	2019	December	18.5

Genres (2019)

```
In [13]: genres_2019 = (  
    tables_2019[2]  
    .loc[5:21] # drop first 5 and last 2 rows  
    .drop(columns=[2, 4])  
)  
  
genres_2019.insert(0, "Year", 2019)  
  
genres_2019.columns = ["Year", "Genre", "Number of releases", "Gross box office (£  
genres_2019.head(7)
```

Out[13]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
5	2019	Action	102	328.3	Avengers: Endgame
6	2019	Animation	40	312.1	The Lion King
7	2019	Drama	264	166.4	Downton Abbey
8	2019	Comedy	136	83.5	Yesterday
9	2019	Biopic	19	80.7	Rocketman
10	2019	Thriller	20	75.3	Joker
11	2019	Sci-fi	3	64.3	Star Wars: The Rise of Skywalker

In [14]: `genres_2019.tail()`

Out[14]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
17	2019	Documentary	99	7.8	Apollo 11
18	2019	Crime	11	2.8	21 Bridges
19	2019	Mystery	1	0.5	The Souvenir
20	2019	War	2	0.1	Saving Private Ryan (D-Day 75th Anniversary)
21	2019	Fantasy	2	<0.1	Gwen

This code is very similar to the one for 2018. We *could* write a generic function to extract and transform a genre-level breakdown from any PDF, but we cannot assume they will all take the same form.

We generally don't want to assume data across multiple files will have the same format, but this is particularly true for PDF reports spanning many years.

Distributors (2019)

```
In [15]: distributors_2019 = (
    tables_2019[4]
    .loc[4:]
    .drop(index=[14, 16, 17]) # drop totals EXCEPT the row totalling "others"
)

distributors_2019.insert(0, "Year", 2019)

distributors_2019.columns = ["Year", "Distributor", "Market share", "Films on release"]
distributors_2019.head()
```

Out[15]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
4	2019	Walt Disney	37.9	24	506.8
5	2019	Universal	13.9	39	185.6
6	2019	Warner Bros	12.4	40	166.3
7	2019	Sony	9.3	27	124.7
8	2019	Paramount	5.9	13	79.0

In [16]: `distributors_2019.tail()`

Out[16]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
10	2019	Lionsgate	4.2	19	56.4
11	2019	Entertainment One	3.8	27	50.4
12	2019	STX Entertainment	1.2	7	15.4
13	2019	StudioCanal	1.0	22	14.0
15	2019	Others (Total 130 distributors)	4.6	671	61.6

2020

In [17]: `tables_2020 = extract_tables("./files/2021 - BFI Statistical Yearbook 2021 - 24979.`

Reading ./files/2021 - BFI Statistical Yearbook 2021 - 24979.pdf
4 tables found.

```
#####
      Table 0
#####
```


	0	1	2	3	4	5	6
0	Table 2 Monthly UK cinema admissions, 2016-2020						
1	Admissions	2016	2017	2018	2019	2020	% change
2	(million)						on 2019
3	January	14.0	15.0	16.2	13.7	16.5	20.4
4	February	15.4	16.5	16.1	12.2	14.5	19.0
5	March	13.4	16.2	13.5	11.4	4.8	-57.9
6	April	13.1	15.6	15.5	16.0	-	-100.0
7	May	12.5	11.3	13.7	16.6	-	-100.0
8	June	10.7	9.6	10.4	13.9	-	-100.0
9	July	16.6	17.8	15.6	18.7	0.4	-97.9
10	August	18.1	14.5	19.2	15.6	2.1	-86.7
11	September	11.7	10.8	10.1	11.0	2.7	-75.8
12	October	15.2	12.1	16.1	16.3	2.2	-86.6
13	November	12.4	14.1	14.9	12.2	0.3	-97.5
14	December	15.2	17.2	15.7	18.5	0.5	-97.2
15	Total	168.3	170.6	177.0	176.1	44.0	-75.0
16	Source: CAA, Comscore						

Table 1
#####

	0	1	2	3	4	5	6
0	Gentlemen, and The Personal History of David C...						
1	Table 10 Films released in the UK and Republic...						
2	Genre	Number of	% of	Gross box	% of total	Average number	Top performing title
3		releases	releases	office	box office	of sites at widest	
4				(£ million)		point of release	
5	Action	36	9.4%	48.5	19.7%	167	Bad Boys For Life
6	War	3	0.8%	44.3	17.9%	296	1917
7	Animation	19	5.0%	34.8	14.1%	260	Sonic The Hedgehog
8	Thriller	15	3.9%	32.8	13.3%	192	Tenet
9	Drama	140	36.7%	31.4	12.7%	83	The Personal History of
10							David Copperfield
11	Comedy	65	17.1%	24.0	9.7%	108	Jojo Rabbit
12	Crime	8	2.1%	12.5	5.1%	118	The Gentlemen
13	Horror	24	6.3%	12.1	4.9%	172	The Invisible Man
14	Adventure	9	2.4%	3.8	1.6%	152	The Call Of The Wild
15	Fantasy	2	0.5%	0.9	0.4%	315	Pinocchio
16	Romance	6	1.6%	0.4	0.2%	30	Hope Gap
17	Documentary	42	11.0%	0.4	0.2%	19	I Am Greta
18	Western	2	0.5%	0.3	0.1%	53	True History Of The Kelly Gang

	0	1	2	3	4	5	6
19	Family	3	0.8%	0.3	0.1%	85	Max Winslow And The
20							House Of Secrets
21	Biopic	3	0.8%	0.2	0.1%	16	Mr. Jones
22	Sci fi	4	1.0%	0.1	0.0%	18	Color Out Of Space
23	Total	381	100.0%	246.9	100.0%	110	

24 Source: Comscore, BFI RSU analysis

Table 2
#####

	0	1	2	3	4
0	Table 1 Distributor share of box office, UK an...				
1	Distributor	Market share	Number of films	Box office	Number of films
2		in 2020	on release	gross in 2020	on release
3		(%)	in 2020	(£ million)	in 2019
4	Entertainment One	15.9	30	49.4	28
5	Sony	15.6	38	48.4	39
6	Walt Disney	14.3	58	44.3	47
7	Universal	12.5	73	39.0	55
8	Warner Bros	11.7	89	36.5	64
9	Paramount	8.0	22	25.0	13
10	Lionsgate	4.5	29	14.0	23
11	StudioCanal	4.5	46	13.9	31
12	Entertainment Film Distributors	4.3	6	13.3	10
13	Shear Entertainment*	1.3	1	4.0	n/a
14	Top 10	92.5	392	287.8	310
15	Other distributors (143)	7.5	534	23.2	844
16	Total	100.0	926	311.0	1,154
17	Source: Comscore				

Table 3
#####

	0	1	2	3	4	5	6	7	8	9	10
0	4%, however, in 2020 this increased to 7.5%, t...										
1	Table 2 Distributor market share as percentage...										
2	Distributor	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
3	Entertainment One	5.1	6.7	9.0	8.0	3.9	8.4	2.5	3.2	3.8	15.9
4	Sony	7.2	18.0	8.7	6.2	11.8	6.6	10.3	10.7	9.3	15.6
5	Walt Disney	8.7	10.2	15.2	10.1	20.0	23.2	19.7	23.6	37.9	14.3
6	Universal	11.8	10.7	15.1	11.2	21.6	14.0	16.0	19.5	13.9	12.5
7	Warner Bros	18.2	12.9	17.2	15.9	9.0	15.6	16.6	13.9	12.4	11.7
8	Paramount	16.3	7.7	7.8	5.8	4.0	5.4	3.7	4.8	5.9	8.0
9	Lionsgate	-	5.7	4.7	5.5	4.0	4.0	6.3	1.5	4.2	4.5
10	Optimum/StudioCanal1	3.8	-	2.8	6.7	4.7	1.5	4.2	2.8	1.0	4.5
11	Entertainment Film Distributors	6.7	3.1	1.9	5.2	1.6	1.5	1.7	1.1	-	4.3
12	Shear Entertainment	-	-	-	-	-	-	-	-	-	1.3
13	20th Century Fox2	12.1	16.1	13.1	21.8	14.9	15.7	15.1	14.5	5.7	-
14	STX Entertainment	-	-	-	-	-	-	-	-	1.2	-
15	Momentum3	4.6	4.3	-	-	-	-	-	-	-	-
16	Top 10 total4	94.4	95.4	95.5	96.3	95.5	95.9	96.1	95.5	95.4	92.5
17	Others	5.6	4.6	4.5	3.7	4.5	4.1	3.9	4.5	4.6	7.5
18	Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
19	Source: Comscore										

```
In [18]: admissions_2020 = (
          tables_2020[0]
          .loc[3:14,[0,5]]
        )

admissions_2020.columns = ["Month", "Admissions (million)"]
admissions_2020.insert(0, "Year", 2020)

admissions_2020.head()
```

```
Out[18]:
```

	Year	Month	Admissions (million)
3	2020	January	16.5
4	2020	February	14.5
5	2020	March	4.8
6	2020	April	-
7	2020	May	-

Let's replace those hyphens with explicit missing (NaN) values

```
In [19]: admissions_2020 = admissions_2020.replace("-", np.NaN)
admissions_2020.head()
```

```
Out[19]:
```

	Year	Month	Admissions (million)
3	2020	January	16.5
4	2020	February	14.5
5	2020	March	4.8
6	2020	April	NaN
7	2020	May	NaN

```
In [20]: admissions_2020.tail()
```

```
Out[20]:
```

	Year	Month	Admissions (million)
10	2020	August	2.1
11	2020	September	2.7
12	2020	October	2.2
13	2020	November	0.3
14	2020	December	0.5

Genres

```
In [21]: genres_2020 = (
    tables_2020[1]
    .loc[5:22] # drop first 5 and last 2 rows
    .drop(columns=[2, 4, 5])
)

genres_2020.insert(0, "Year", 2020)

genres_2020.columns = ["Year", "Genre", "Number of releases", "Gross box office (£
genres_2020.head(8)
```

Out[21]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
5	2020	Action	36	48.5	Bad Boys For Life
6	2020	War	3	44.3	1917
7	2020	Animation	19	34.8	Sonic The Hedgehog
8	2020	Thriller	15	32.8	Tenet
9	2020	Drama	140	31.4	The Personal History of
10	2020				David Copperfield
11	2020	Comedy	65	24.0	Jojo Rabbit
12	2020	Crime	8	12.5	The Gentlemen

In [22]: `genres_2020.tail()`

Out[22]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
18	2020	Western	2	0.3	True History Of The Kelly Gang
19	2020	Family	3	0.3	Max Winslow And The
20	2020				House Of Secrets
21	2020	Biopic	3	0.2	Mr. Jones
22	2020	Sci fi	4	0.1	Color Out Of Space

Looks like a data error on row 10 and 20 - we might encounter a few of these where table rows spill over.

Let's manually fix these since we're here.

```
In [23]: genres_2020.loc[9, "Top performing title"] = "The Personal History of David Copperf
genres_2020.loc[19, "Top performing title"] = "Max Winslow And The House Of Secrets

genres_2020 = genres_2020.drop(index=[10, 20])
genres_2020.head()
```

Out[23]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
5	2020	Action	36	48.5	Bad Boys For Life
6	2020	War	3	44.3	1917
7	2020	Animation	19	34.8	Sonic The Hedgehog
8	2020	Thriller	15	32.8	Tenet
9	2020	Drama	140	31.4	The Personal History of David Copperfield

In [24]: `genres_2020.tail()`

Out[24]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
17	2020	Documentary	42	0.4	I Am Greta
18	2020	Western	2	0.3	True History Of The Kelly Gang
19	2020	Family	3	0.3	Max Winslow And The House Of Secrets
21	2020	Biopic	3	0.2	Mr. Jones
22	2020	Sci fi	4	0.1	Color Out Of Space

Distributors

In [25]: `tables_2020[2]`

Out[25]:

	0	1	2	3	4
0	Table 1 Distributor share of box office, UK an...				
1	Distributor	Market share	Number of films	Box office	Number of films
2		in 2020	on release	gross in 2020	on release
3		(%)	in 2020	(£ million)	in 2019
4	Entertainment One	15.9	30	49.4	28
5	Sony	15.6	38	48.4	39
6	Walt Disney	14.3	58	44.3	47
7	Universal	12.5	73	39.0	55
8	Warner Bros	11.7	89	36.5	64
9	Paramount	8.0	22	25.0	13
10	Lionsgate	4.5	29	14.0	23
11	StudioCanal	4.5	46	13.9	31
12	Entertainment Film Distributors	4.3	6	13.3	10
13	Shear Entertainment*	1.3	1	4.0	n/a
14	Top 10	92.5	392	287.8	310
15	Other distributors (143)	7.5	534	23.2	844
16	Total	100.0	926	311.0	1,154
17	Source: Comscore				

```

In [26]: distributors_2020 = (
    tables_2020[2]
    .loc[4:15,:3]
    .drop(index=[14]) # drop totals EXCEPT the row totalling "others"
)

distributors_2020.insert(0, "Year", 2020)

distributors_2020.columns = ["Year", "Distributor", "Market share", "Films on relea
distributors_2020.head()

```

Out[26]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
4	2020	Entertainment One	15.9	30	49.4
5	2020	Sony	15.6	38	48.4
6	2020	Walt Disney	14.3	58	44.3
7	2020	Universal	12.5	73	39.0
8	2020	Warner Bros	11.7	89	36.5

In [27]: `distributors_2020.tail()`

Out[27]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
10	2020	Lionsgate	4.5	29	14.0
11	2020	StudioCanal	4.5	46	13.9
12	2020	Entertainment Film Distributors	4.3	6	13.3
13	2020	Shear Entertainment*	1.3	1	4.0
15	2020	Other distributors (143)	7.5	534	23.2

2021

In [28]: `tables_2021 = extract_tables('./files/2022 - The box office 2021 - full report - 26`

Reading ./files/2022 - The box office 2021 - full report - 26781.pdf
2 tables found.

```
#####
      Table 0
#####
```

0	1
0	Shang-Chi's three-day record was soon overtake... At 8.8 million, attendances in November were a...
1	debut of another official 25th outing. The muc... the level of October but were still the fourth...
2	007 instalment, No Time to Die, originally sch... year. Releases in the month include three top ...
3	release in April 2020, grossed £21 million on ... – Ghostbusters Afterlife and the UK studio-bac...
4	opening (£25.8 million including Thursday prev... Eternals and House of Gucci – as well as the t...
5	highest Friday to Sunday tally for a Bond film... grossing UK independent film of the year, Spen...
6	UK and Republic of Ireland. Officially release... Admissions rose again in December, despite con...
7	1 October, No Time to Die would go on to becom... around the discovery of the COVID Omicron vari...
8	UK's third highest ever release with a gross o... biggest release of the month was Spider-Man: N...
9	and was the first film of the year to entice s... Home, which would go on to become the second h...
10	numbers of older audience members back to the ... grossing release of the year and the all-time ...
11	Alongside three other top 20 earners – Dune, t... highest earning film at the UK and Republic of...
12	studio-backed title Venom: Let There Be Carnag... box office with earnings (to 13 February 2022) of
13	The Addams Family 2 – and the top earning UK £93 million. Ticket sales for the month were a...
14	independent film of the year, The French Dispa... second highest of the year, with total admissi...
15	Time to Die helped propel October's ticket sal... 13.5 million.
16	highest total of the year at 16.4 million. Oct...
17	only month to see an increase in admissions co...
18	with the equivalent month in 2019.
19	Table 2 Monthly UK cinema admissions, 2019-2021

#####

Table 1

#####

	0	1	2	3	4
0	Table 2 Monthly UK cinema admissions, 2019-2021				
1	Admissions (million)	2019	2020	2021	% change on 2019
2	January	13.7	16.5	-	-100.0
3	February	12.2	14.5	-	-100.0
4	March	11.4	4.8	-	-100.0
5	April	16.0	-	-	-100.0
6	May	16.6	-	3.5	-78.8
7	June	13.9	-	7.0	-49.9
8	July	18.7	0.4	7.8	-58.5
9	August	15.6	2.1	10.4	-33.1
10	September	11.0	2.7	6.5	-40.6
11	October	16.3	2.2	16.4	0.9
12	November	12.2	0.3	8.8	-27.5
13	December	18.5	0.5	13.5	-26.8
14	Total	176.1	44.0	74.0	-58.0
15	Source: CAA, Comscore				

Looks like Table 8 is monthly UK cinema admissions

```
In [29]: admissions_2021 = (  
    tables_2021[1]  
    .loc[2:13]  
    .drop(columns=[1, 2, 4])  
)  
  
admissions_2021.columns = ["Month", "Admissions (million)"]  
admissions_2021.insert(0, "Year", 2021)  
  
admissions_2021["Admissions (million)"] = admissions_2021["Admissions (million)"].r  
admissions_2021.head()
```

Out[29]:

	Year	Month	Admissions (million)
2	2021	January	NaN
3	2021	February	NaN
4	2021	March	NaN
5	2021	April	NaN
6	2021	May	3.5

In [30]: `admissions_2021.tail()`

Out[30]:

	Year	Month	Admissions (million)
9	2021	August	10.4
10	2021	September	6.5
11	2021	October	16.4
12	2021	November	8.8
13	2021	December	13.5

Genres

For the first time these stats are actually in another PDF

In [31]: `tables_2021_genre = extract_tables("./files/2022 - Top films in 2021 - full report`

Reading ./files/2022 - Top films in 2021 - full report - 29763.pdf
1 tables found.

```
#####
      Table 0
#####
```


	0	1	2	3	4	5	6
20							Not Be Televised)
21	Romance	13	2.9	1.5	0.2	77	The Last Letter from Your Lover
22	Crime	2	0.5	0.3	<0.1	137	Rise of the Footsoldier: Origins
23	Total	442	100.0	616.2	100.0	138	
24	Source: Comscore, BFI RSU analysis						

```
In [32]: genres_2021 = (
    tables_2021_genre[0]
    .loc[5:22]
    .drop(columns=[2, 4, 5])
)

genres_2021.insert(0, "Year", 2021)

genres_2021.columns = ["Year", "Genre", "Number of releases", "Gross box office (£

# fix data error
genres_2021.loc[18, "Top performing title"] = "Summer of Soul (...Or, When the Revo
genres_2021 = genres_2021.drop(index=[19, 20])

genres_2021.head()
```

```
Out[32]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
5	2021	Action	52	296.7	No Time to Die
6	2021	Comedy	54	65.1	Free Guy
7	2021	Animation	38	58.2	The Addams Family 2
8	2021	Horror	24	40.8	A Quiet Place Part II
9	2021	Family	8	29.6	Peter Rabbit 2: The Runaway

```
In [33]: genres_2021.tail()
```

Out[33]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
16	2021	Thriller	10	8.2	The Forever Purge
17	2021	Suspense	12	4.5	Old
18	2021	Documentary	69	2.1	Summer of Soul (...Or, When the Revolution Cou...
21	2021	Romance	13	1.5	The Last Letter from Your Lover
22	2021	Crime	2	0.3	Rise of the Footsoldier: Origins

Distributors

In [34]: `tables_2021_dist = extract_tables("./files/2022 - Distribution - full report - 2678`

Reading ./files/2022 - Distribution - full report - 26782.pdf
2 tables found.

```
#####  
      Table 0  
#####
```


0	Table 1 shows box office takings by distributo...	
1	Distributors	
2	films on release in the UK and Republic of Ire...	
3	2021 (including titles released prior to 2021)...	
4	earning distributor was Universal which releas...	
5	A strong recovery in the numbers of people cho...	highest earning film of the year No Time to Di...
6	watch films on the big screen made 2021 a more	four other top 20 titles including Fast & Furi...
7	successful year for the distribution sector th...	Addams Family 2. In total, titles released by ...
8	Despite continued challenges from the COVID-19	generated revenues of £181 million in 2021 (a ...
9	pandemic, which saw all cinemas closed for the...	of overall box office). The second highest ear...
10	to six months of the year, a total of 981 film...	distributor in 2021 was Sony which released fo...
11	release theatrically during 2021, compared wit...	year's top 20 titles, including Spider-Man: No...
12	in 2020 (1,154 in 2019). Cinemas reopened in E...	Peter Rabbit 2: The Runaway and Ghostbusters: ...
13	Scotland and Wales from 17 May, in Northern Ir...	Sony releases earned £131 million in 2021, a 2...
14	from 24 May and in the Republic of Ireland fro...	of the box office.
15	The top 10 feature film distributors had a 97%...	The appearance in the top 10 of Park Circus, w...
16	the market in 2021 from the release of 432 tit...	specialises in classic and back catalogue titl...
17	all films on release). The share is the highes...	the importance of older releases to cinemas du...
18	and up almost five percentage points from 2020...	two years most impacted by the pandemic. The
19	the top 10 distributors had a market share of ...	newcomer to the list had 96 titles on release ...
20	392 releases (42% of all releases). A further 137	the greatest number of any of the top 10 distr...
21	distributors were involved in the theatrical r...	

22 in the UK and Republic of Ireland in 2021,
com...

23 143 companies outside the top 10 in 2020.
These

24 distributors handled a total of 549 titles (56...

25 releases) but generated only 3% of the total
b...

26 Table 1 Distributor share of box office, UK an...

#####

Table 1

#####

	0	1	2	3
0	Table 1 Distributor share of box office, UK an...			
1	Distributor	Market share in 2021 (%)	Films on release in 2021	Box office gross in 2021
2				(£ million)
3	Universal	30.6	51	180.8
4	Sony	22.2	52	130.8
5	Walt Disney	21.2	63	125.1
6	Warner Bros	14.4	80	84.7
7	Paramount	3.7	16	21.8
8	Lionsgate	1.8	23	10.5
9	Entertainment One	1.4	12	8.1
10	Park Circus	0.6	96	3.4
11	STX Entertainment	0.6	3	3.4
12	StudioCanal	0.5	36	2.7
13	Top 10	96.8	432	571.2
14	Other distributors (137)	3.2	549	19.0
15	Total	100.0	981	590.2
16	Source: Comscore			

```
In [35]: distributors_2021 = (
    tables_2021_dist[1]
    .loc[3:14]
    .drop(index=[13]) # drop Top 10 total row
```

```
)

distributors_2021.insert(0, "Year", 2021)
distributors_2021.columns = ["Year", "Distributor", "Market share", "Films on release", "Box office gross (£ million)"]
distributors_2021.head()
```

```
Out[35]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
3	2021	Universal	30.6	51	180.8
4	2021	Sony	22.2	52	130.8
5	2021	Walt Disney	21.2	63	125.1
6	2021	Warner Bros	14.4	80	84.7
7	2021	Paramount	3.7	16	21.8

```
In [36]: distributors_2021.tail()
```

```
Out[36]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
9	2021	Entertainment One	1.4	12	8.1
10	2021	Park Circus	0.6	96	3.4
11	2021	STX Entertainment	0.6	3	3.4
12	2021	StudioCanal	0.5	36	2.7
14	2021	Other distributors (137)	3.2	549	19.0

Combine everything

```
In [37]: admissions = pd.concat([admissions_2018, admissions_2019, admissions_2020, admissions_2021],
                                ignore_index=True,
                                axis=0)

admissions.head()
```

```
Out[37]:
```

	Year	Month	Admissions (million)
0	2018	January	16.2
1	2018	February	16.1
2	2018	March	13.5
3	2018	April	15.5
4	2018	May	13.7

```
In [38]: admissions["Year"].value_counts()
```

```
Out[38]: 2018    12
         2019    12
         2020    12
         2021    12
         Name: Year, dtype: int64
```

```
In [39]: admissions.to_csv("admissions.csv", index=False)
```

Genres

```
In [40]: genres = pd.concat([genres_2018, genres_2019, genres_2020, genres_2021],
                             ignore_index=True,
                             axis=0)

genres.head()
```

```
Out[40]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2018	Action	77	361.3	Avengers: Infinity War
1	2018	Animation	46	242.2	Incredibles 2
2	2018	Drama	246	154.1	A Star Is Born
3	2018	Comedy	144	88.3	Johnny English Strikes Again
4	2018	Biopic	5	78.5	Bohemian Rhapsody

```
In [41]: genres.tail()
```

```
Out[41]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
62	2021	Thriller	10	8.2	The Forever Purge
63	2021	Suspense	12	4.5	Old
64	2021	Documentary	69	2.1	Summer of Soul (...Or, When the Revolution Cou...
65	2021	Romance	13	1.5	The Last Letter from Your Lover
66	2021	Crime	2	0.3	Rise of the Footsoldier: Origins

```
In [42]: genres["Year"].value_counts()
```

```
Out[42]: 2018    18
         2019    17
         2020    16
         2021    16
         Name: Year, dtype: int64
```

```
In [43]: genres.to_csv("genres.csv", index=False)
```

Distributors

```
In [44]: distributors = pd.concat([distributors_2018, distributors_2019, distributors_2020,  
                                ignore_index=True,  
                                axis=0)  
  
distributors.head()
```

```
Out[44]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2018	Walt Disney	23.6	24	325.6
1	2018	Universal	19.5	40	268.5
2	2018	20th Century Fox	14.5	28	199.3
3	2018	Warner Bros	13.9	31	191.4
4	2018	Sony	10.7	33	146.8

```
In [45]: distributors.tail()
```

```
Out[45]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
39	2021	Entertainment One	1.4	12	8.1
40	2021	Park Circus	0.6	96	3.4
41	2021	STX Entertainment	0.6	3	3.4
42	2021	StudioCanal	0.5	36	2.7
43	2021	Other distributors (137)	3.2	549	19.0

```
In [46]: distributors["Year"].value_counts()
```

```
Out[46]: 2018    11  
         2019    11  
         2020    11  
         2021    11  
         Name: Year, dtype: int64
```

```
In [47]: distributors.to_csv("distributors.csv", index=False)
```